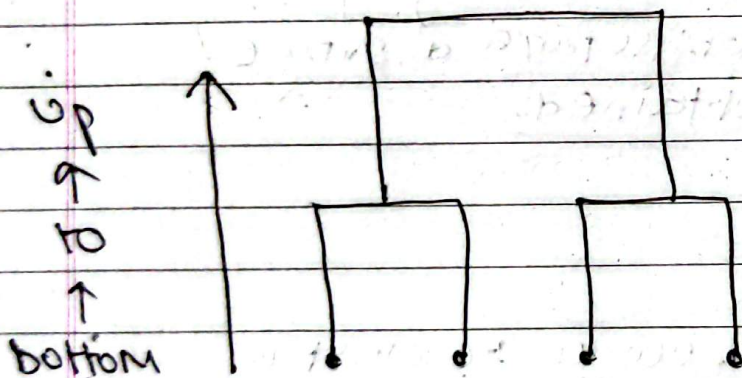


# Agglomerative Hierarchical clustering

CLIFF

- \* **Definition:** Agglomerative is a bottom up approach in which the algorithm starts with taking all data points as single cluster and merging them until one cluster is left.

## Diagram



## \* Why we need This?

- 1) using dendrogram, only the groups of data point can be shown, no need of  $k$  here.
- 2) it works with distance between points.
- 3) Best for Numerical data.
- 4) used only for maximum 1000 data points.



## Algorithm of Agglomerative clustering

- ① Start by considering each data point as a separate cluster (group).
- ② calculate distance between every pair of cluster.
- ③ Find the two closest cluster and joined them making one.
- ④ update distance matrix after merging.
- ⑤ repeat step 3 and 4 until reach a single / only one cluster is obtained.

How do we measure distance to cluster distance?

① Single linkage

→ distance between closest pair of points, one from each cluster.

$$d(A, B) = \min d(a, b)$$

$$a \in A, b \in B$$

Finds elongated, chain like clusters.



- (2) Complete linkage  
 → Distance between the furthest pair of points.

$$d(A, B) = \max_{a \in A, b \in B} d(a, b)$$

produces compact, similar sized clusters.

- (3) Average linkage  
 → Average distance over all cross cluster pairs.

$$d(A, B) = \frac{1}{|A| |B|} \sum_{a \in A} \sum_{b \in B} d(a, b)$$

A compromise between the two.

### Numerical Example

| Point          | x    | y    |
|----------------|------|------|
| P <sub>1</sub> | 0.40 | 0.53 |
| P <sub>2</sub> | 0.22 | 0.38 |
| P <sub>3</sub> | 0.35 | 0.32 |
| P <sub>4</sub> | 0.26 | 0.19 |
| P <sub>5</sub> | 0.08 | 0.41 |
| P <sub>6</sub> | 0.45 | 0.30 |



Step 1) calculate distance matrix

$$d(P_1, P_2) = \sqrt{(0.40 - 0.22)^2 + (0.53 - 0.38)^2} \\ \approx 0.23$$

Doing this for all points pairs (value rounded to 2 decimal places)

|       | $P_1$ | $P_2$ | $P_3$ | $P_4$ | $P_5$ | $P_6$ |
|-------|-------|-------|-------|-------|-------|-------|
| $P_1$ | 0     |       |       |       |       |       |
| $P_2$ | 0.23  | 0     |       |       |       |       |
| $P_3$ | 0.22  | 0.15  | 0     |       |       |       |
| $P_4$ | 0.37  | 0.20  | 0.15  | 0     |       |       |
| $P_5$ | 0.34  | 0.14  | 0.28  | 0.29  | 0     |       |
| $P_6$ | 0.23  | 0.25  | 0.11  | 0.22  | 0.39  | 0     |

Step observe the smallest distance point in table which is 0.11 among all of  $P_6$  corresponding to  $P_3$ .

So, merging  $P_3$  to  $P_6$

Step 2: merging  $\{P_3, P_6\}$

|            | $P_1$ | $P_2$ | $P_3, P_6$ | $P_4$ | $P_5$ |
|------------|-------|-------|------------|-------|-------|
| $P_1$      | 0     |       |            |       |       |
| $P_2$      | 0.23  | 0     |            |       |       |
| $P_3, P_6$ | 0.22  | 0.15  | 0          |       |       |
| $P_4$      | 0.37  | 0.20  | 0.15       | 0     |       |
| $P_5$      | 0.34  | 0.14  | 0.28       | 0.29  | 0     |



Next smallest distance is 0.14 of  $P_5$  corresponding to  $P_2$

Merging  $\{P_2, P_5\}$

Step 3: Merge  $\{P_2, P_5\}$

|            | $P_1$ | $P_2, P_5$ | $P_3, P_6$ | $P_4$ |
|------------|-------|------------|------------|-------|
| $P_1$      | 0     |            |            |       |
| $P_2, P_5$ | 0.23  | 0          |            |       |
| $P_3, P_6$ | 0.22  | 0.15       | 0          |       |
| $P_4$      | 0.37  | 0.20       | 0.15       | 0     |

Next smallest are two so, first taking one 0.15 of  $P_3, P_6$  corresponding to  $P_2, P_5$

Merging  $\{P_3, P_6, P_2, P_5\}$

Step 4: Merge  $\{P_2, P_3, P_5, P_6\}$

|                      | $P_1$ | $P_2, P_3, P_5, P_6$ | $P_4$ |
|----------------------|-------|----------------------|-------|
| $P_1$                | 0     |                      |       |
| $P_2, P_3, P_5, P_6$ | 0.22  | 0                    |       |
| $P_4$                | 0.37  | 0.20                 | 0     |

Next smallest is 0.20 of  $P_4$  corresponding to  $P_2, P_3, P_5, P_6$

$P_2, P_3, P_5, P_6$

Merging  $P_2, P_3, P_4, P_5, P_6$



Step 8 Merge  $P_2, P_3, P_4, P_5, P_6$

|                           |       |                           |
|---------------------------|-------|---------------------------|
|                           | $P_1$ | $P_2, P_3, P_4, P_5, P_6$ |
| $P_1$                     | 0     |                           |
| $P_2, P_3, P_4, P_5, P_6$ | 0.22  | 0                         |

Now final  $P_1$  remains with 0.22 distance

Step 8 One big cluster is gone

→  $P_1$  joins at distance 0.22

→ All six points are now one cluster ⇒ the algorithm stops

Dendrogram

